

Entropic Tug of War: Topological Constraints Spontaneously Rectify the Dynamics of a Polymer with Heterogeneous Fluctuations

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Polymers with active segments constitute prospective future materials and are used as a model for some biological systems such as chromatin. The directions of the active forces are typically introduced with temporal or spatial correlations to establish directional motion of the chain and corresponding active dynamics. Instead, here we consider an active-passive copolymer, where the two segments differ only by the magnitude of their fluctuations and feature no artificial correlations. Here we show that although the model itself does not possess directional dynamics, if the chains are concentrated, directional persistent motion spontaneously arises as a consequence of the broken translational symmetry owing to the topological constraints. Using scaling arguments and simulations, we explain the phenomenon and describe the ensuing dynamics. Our work has thus far-reaching consequences for the mechanical properties of all dense active polymeric systems with heterogeneous fluctuations and in particular for chromatin conformation and dynamics that are crucial for biological functionality.

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I. INTRODUCTION

Active polymers, encountered in nature as chromatin fiber, cytoskeleton, or living worms [1–3], form a subclass of active matter characterized by out-of-equilibrium driving on the single filament level, sharing many of the collective phenomena found in colloidal active matter [4,5]. Besides biology, activity seems a promising control parameter for the tuning of the constituent material properties. In particular, the polymers feature topological constraints, because the chains cannot cross each other, that, as a result, in passive systems govern the stress relaxation and their viscoelastic properties [6]. While for active, *stiff* filaments the uncrossability in mutual collisions generates torques and ordering, for *flexible* polymers that we focus on in here, it allows for multichain entanglements affecting the material properties in various contrasting ways. The energy input owing to the activity typically speeds up the (entanglement) relaxation, due to the active propagation of the chain along its contour, thereby decreasing the apparent viscosity [7–9]. In contrast, the activity can create new forms of entanglements, inaccessible in equilibrium,

generating novel phases of matter, such as the active topological glass [10–12].

The directional and persistent motion of the chains underlying both contrasting trends in material properties is typically introduced in the microscopic model by hand, e.g., by imposing the active force or velocity to act in the backbone direction [8,9,13–15], requiring a substrate with fixed motor proteins [16], or composing the polymer of active Brownian or Ornstein-Uhlenbeck particles that have the persistence built in [17–19]. Here we show, on the contrary, that the active polymer persistence on the chain scale can arise *spontaneously* from isotropic, nonequilibrium, thermal-like fluctuations due to the presence of the topological constraints. Using scaling arguments, supported by simulations, we uncover that the topological restriction of the transverse relaxation modes results in an imbalance of forces of entropic origin at the chain ends, which generates the directional dynamics.

The activity mechanism we use has no persistence and is isotropic. We model it, on each monomer of a contiguous segment of the chain, as thermal-like fluctuations (delta-correlated white noise force with no correlation between the monomers) but with larger amplitude (temperature) than the rest of the chain. Therefore, the active nonequilibrium character of the chain stems from the fact that distinct segments are maintained at distinct temperatures. This “scalar” activity model is not only interesting from the perspective of materials science, where the stronger fluctuations can be induced by monomer-specific external fields, but also for its relation to chromatin dynamics.

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The active processes of chromatin (remodelling, transcription, loop extrusion, transcription factor binding, etc.) are mediated by the corresponding molecular machinery and exert nonequilibrium fluctuating forces on specific sections of the chromatin fiber [20]. Some of the active processes manifest themselves by stronger thermal-like fluctuations, while others have more complicated spectra [21,22]. In general the activity impacts the chromatin dynamics [21,23] and its conformation [24–27], thereby acting on the biological functionality of the cell [20]. Although the role of active fluctuations in chromatin folding is not yet settled, the topological constraints have been shown to govern its large-scale organization [28]. Our present work indicates that the explicit consideration of both the active processes and the chain noncrossability can be crucial in interpreting the chromatin dynamics, as in synergy with the activity the constraints rectify the chains' motion.

The equations of motion for such a polymer are coupled Langevin equations for all the N monomers

$$m\ddot{\mathbf{r}}_i = -\gamma\dot{\mathbf{r}}_i - \nabla_{\mathbf{r}_i} U + \sqrt{6k_B\gamma T_i}\boldsymbol{\eta}_i, \quad (1)$$

with $\boldsymbol{\eta}_i$ being uncorrelated random force of zero mean and unit variance. The corresponding noise term amplitude acting on monomer i is governed by the temperature T_i that is T_h , for the monomers belonging to the hot segment or T_c for the ones that make the cold segment. In free space and in the long time limit, the center of mass of the polymer exhibits equilibriumlike diffusive dynamics as can be seen by summing up all the Langevin equations, giving

$$M\ddot{\mathbf{r}}_{\text{cm}} = -\Gamma\dot{\mathbf{r}}_{\text{cm}} + \sqrt{6k_B\Gamma} \sum_i^N (T_i/N)^{1/2} \boldsymbol{\eta}_i. \quad (2)$$

The potential forces from the monomers ($-\nabla_{\mathbf{r}_i} U$) add up to zero, $M = Nm$ is the total mass and $\Gamma = N\gamma$ the total friction coefficient. The random force term can be written also as $\sqrt{6k_B\Gamma T_{\text{eff}}}\boldsymbol{\eta}$, where the effective temperature $T_{\text{eff}} = (T_h N_h + T_c N_c)/(N_c + N_h)$ is the average temperature weighted by the number of monomers $N_{h(c)}$ in the hot (cold) segments and $\boldsymbol{\eta}$ is a zero mean and unit variance random variable. Note, however, that such a form obscures the fact that the equation for the center of mass is correlated to the equations governing the internal degrees of freedom through the noise term. The simplest way to see this, is in the special case of a dumbbell with two different temperatures, as we show in Appendix Sec. B 1. The result of the correlation is twofold. First, the long-time equilibriumlike dynamics of the dumbbell is characterized by a diffusion coefficient governed by the combination of T_h and T_c other than T_{eff} as shown in [29]. Second, there exists a correlation of the center of mass motion in the direction of the hot bead that occurs on a timescale below the relaxation time of the dumbbell, as shown, e.g., in [10]. The short-time

directional motion of the center of mass can also arise in the case of a dumbbell at a single temperature, but different friction coefficients of the two beads [30]. The center of mass can also exhibit a genuine drift in the long-time limit, but in a more general model of a dumbbell, the so-called Brownian inchworm [31], as a result of a stretching-dependent friction, which is not present in our model.

While it is possible to derive the dynamic characteristics of a two-temperature dumbbell, the systems relevant in material and biological contexts involve longer polymers at high densities, making the situation much more complex for several reasons. First, it is the presence of many-body effects that in multitemperature systems can challenge the equilibrium analogies [32,33]. Second, the presence of excluded volume effects and, importantly, topological constraints, make the previously used techniques based on the Rouse model of a freely crossable chain in free space unsuitable [25]. Interestingly, simulations show [10,11] that the dynamics of a linear chain or a ring with active segment in a melt sharply contrasts with equilibrium dynamics. In the active melt, the hot segment pulls the rest of the chain through the mesh of the other chains, forcing it to move along the delineated curvilinear path. The scaling of mean squared displacement $g_1(t)$ of a tagged passive monomer as a function of time is then determined by the chain statistics, namely, the scaling of the path's end-to-end distance R with its contour coordinate s as $R \sim s^\nu$. For ballistic motion ($s \sim t$) along the path, $g_1(t) \sim R^2[s(t)] \sim t^{2\nu}$ results, as we detail later. Therefore, the monomer dynamics is superdiffusive for $\nu > 1/2$, and the same was also found for the center of mass mean squared displacement $g_3(t)$ of the chains in active topological glass [10]. In contrast, in equilibrium the chain also moves within the effective tube formed by the other uncrossable chains, but only diffusively, which results in the well-known subdiffusive reptation dynamics [34,35]. From a coarse-grained perspective, it is tempting to ascribe the superdiffusive dynamics to the correlated center of mass dynamics found for dumbbells, but as we demonstrate the situation is more complex.

We first show the topological origin of the superdiffusive dynamics. Then we discuss the dynamical consequences of the directionality on the motion of a monomer and the chain's center of mass. We show how the scaling exponents of the mean squared displacement with time depend on the chain parameters and the spacing of the topological constraints. We conclude with the discussion of the relation to other observed effects in two-temperature polymer melts, chromatin and active-copolymer viscoelasticity.

II. RESULTS

A. Entropy imbalance generates polymer drift

Every chain of a concentrated polymer solution or a melt, is confined in an effective tube formed by the other chains in the system. However, even in equilibrium there is a

distinction between a chain confined in a real tube and the chain confined by other chains. The difference lies in the chain's primitive path. The primitive path is the end-to-end contour length modulo the transverse excursions that are not topologically wrapped around the other chains i.e. could be reeled in from the chain ends (see Fig. 7 in the Appendix for illustration). The free energy (of entropic origin) cost to stretch a Gaussian chain to end-to-end distance R scales as $F_s \sim k_B TR^2/2Nb^2$, or equivalently in terms of the primitive path length L as $F_s \sim k_B TL^2/2Nb^2$, b being the monomer size [36]. In free space or confinement the free energy is minimal at zero stretch, while in practice the fluctuations of order $k_B T$ give the random walk statistics $R \sim L \sim bN^{1/2}$. As is well known, in the presence of other chains the situation is different [36]. While the end-to-end distance remains mostly unaffected (Flory screening of the excluded volume), the primitive path length is strongly stretched. A mean-field picture of the melt is that of a chain in a lattice of impenetrable obstacles with some spacing a and coordination number z [Fig. 1(a)]. The effective lattice constant a represents the so-called entanglement tube diameter corresponding to the spatial distance between two entanglements that are restricting the chains' motion. The scale a is sensitive to the density and the bending stiffness of the polymer system [37,38], and it can be much larger than the monomer size or the inter-chain distance because not every contact restricts the chain significantly. At every opening ("gate") between the obstacles (entanglements) the chain end has $z - 1$ options to stretch its primitive path, but only one to return and shorten it. Therefore, this "topological" entropy, originating from the broken translational invariance, favors larger L ,

approximately $F_t \sim -k_B TL/a$, where we dropped factors of order unity involving z . Minimizing the total $F_s + F_t$ gives the standard result of the highly stretched primitive path length, $L \sim b^2 N/a$ [36]. The stretching is the consequence of the entropic force $f = -dF_t/dL \sim k_B T/a$ pulling the chain at both ends. An exact result of the entropy of the primitive path is known [39], showing that the above "harmonic" approximation is accurate.

In a lattice of obstacles, a chain with two distinct temperatures at its ends, experiences entropic forces at the two ends that do not compensate each other, resulting in the force difference

$$\Delta f \sim \frac{k_B \Delta T}{a}, \quad (3)$$

between the chain ends. Although the system is inherently out of equilibrium and the definition of entropy is problematic, the two ends have locally well-defined temperatures and can be described by two competing entropies and the resulting forces. The force imbalance (3) causes a drift velocity $v \sim \Delta f/\Gamma \sim k_B \Delta T/\Gamma a$ of the monomers along the tube delineated by the obstacles and the contour of the chain.

To test the theory we run molecular dynamics simulations of a di-block fully flexible linear chain with identical purely repulsive potentials [40], where the two blocks differ only by the associated temperatures T_h and T_c modeled by random forces with the corresponding distinct amplitudes. See methods section for details of the model and its natural units of length, σ , characterizing the monomer size and time τ . We place the chain in a 3D cubic mesh of impenetrable obstacles, with the lattice constant a [see Fig. 1(a) for a 2D sketch], in the range $3\sigma \leq a \leq 15\sigma$, representing other chains in a melt system. We do so to avoid other, higher-order effects such as the temperature-dependence of the local density or friction, of the number of monomers between entanglements N_e or the presence of active-passive (micro)phase separation [29,41–44]. We aim to measure the dependence of the contour velocity on the temperature contrast ΔT and the lattice spacing a , to compare it with (3).

We begin by equilibrating a polymer chain of $N = 200$ monomers at a temperature $T_c = 1.0$ (see methods). After equilibration, we activate the first $N_h = 24$ monomers by coupling them to a Langevin thermostat at a higher temperature $T_h > T_c$ and keep the rest of the chain with thermostat at T_c . The system is allowed to evolve and we discard the initial period of $10^5 \tau$ until the gyration radius of the chain reaches a steady state (about $5 \times 10^4 \tau$). We checked that the total run of over $1.5 \times 10^6 \tau$ is long enough for the chain to move more than 8 times its steady-state gyration radius and reach a diffusive regime in all the systems.

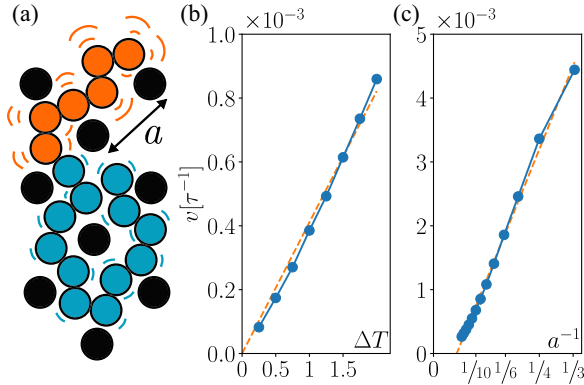


FIG. 1. (a) A 2D sketch of the simulated polymer with a hot segment at temperature T_h (orange) and a cold segment at a temperature T_c (blue) in a lattice of impenetrable static obstacles (black) forming a square mesh with openings of size a and coordination number $z = 4$. (b) and (c) Contour velocity in units of lattice constant per τ as a function of the temperature contrast ΔT (b) for $a = 9$, and as a function of the inverse lattice spacing $1/a$ (c) for $T_h = 3$. Dashed lines are linear fits to the data (symbols).

To analyze the chain's dynamics, we compute the effective tube of the polymer conformation delineating its primitive path. Each monomer is assigned to a resident cell in the lattice, excluding cells associated with distant transverse fluctuations, i.e., when the chain temporarily leaves and returns to a cell without encircling an obstacle (see Fig. 7 in the Appendix). As the chain moves, part of the tube is vacated, and another part is newly created. By comparing the tubes in snapshots separated by Δt we identify the surviving part of the tube and for each monomer from the surviving part we compute the displacement along the tube. Averaging these displacements over multiple snapshots yields the contour displacement profile $d_c(s)$ along the chain, which reveals a steady contour velocity in the middle section of the chain directed toward the hot segment (Fig. 8).

The mean contour velocity v (in the units of number of lattice cells per τ), computed as $v = \langle d_c(s)/\Delta t \rangle_{s \in [75,125]}$, is independent of the different time lags used ($\Delta t = 500\text{--}2000\tau$) (see Fig. 8). Simulations for different $\Delta T = T_h - T_c$ and different a confirm the linear scalings $v \sim \Delta T$ as well as $v \sim a^{-1}$ [Figs. 1(b) and 1(c)]. While in the former case, the linear fit confirms $v(\Delta T = 0) = 0$, the $v \sim a^{-1}$ requires a two-parameter fit

$$v \sim \frac{\Delta T}{a} - \frac{\Delta T}{a_{\max}}, \quad (4)$$

where we find $a_{\max} = 18.2$, which is comparable to the chain size characterized by its gyration radius $R_g \simeq 12.5\sigma$ or the end-to-end distance $R_{ee} \simeq 29.5\sigma$ (both mostly independent on a —see Fig. 9). Clearly, the theoretical result (3) holds when the chain is much larger than the lattice spacing a , while if the chain is smaller than the lattice spacing, there are no topological constraints and we expect $v \rightarrow 0$.

We can verify these results also by computing the diffusion coefficient D of the whole chain. This can be estimated as $D \sim R^2/\tau_{\text{relax}}$, where R^2 is the squared end-to-end chain size and τ_{relax} , the relaxation time, is the time to reach a completely uncorrelated conformation. The relaxation time is the time it takes for the chain to get out of its initial tube as then indeed it renews its conformation and moves about its own size [34]. The length of the primitive path or tube can be also written as $L \sim aN/N_e$, where N_e is the number of monomers per one cell (Fig. 7). To estimate the relaxation time, one has to consider, in general, both the diffusion and the drift processes along the tube. While the diffusion time scales as L^2/D_c , where D_c is the contour diffusion coefficient, the drift time scales as L/v . For long chains, the number of cells forming the primitive path is large, $L/a = N/N_e \gg 1$, and hence the drift relaxation process will be much faster than the contour diffusion of the chain. Therefore, the relaxation time $\tau_{\text{relax}} \sim (N/N_e)/v$. On that timescale, the chain moves its own size $R^2 \sim Nb^2$ in the 3D space, so the diffusion coefficient

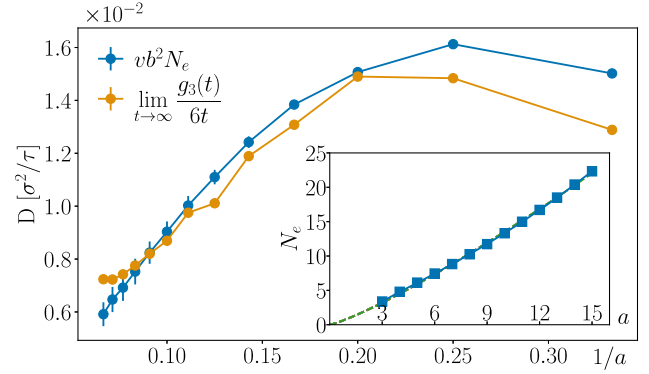


FIG. 2. Diffusion coefficient as a function of the inverse lattice spacing $1/a$ for $T_h = 3.0$ measured from the contour velocity as $v b^2 N_e$ (blue) or directly from the long time mean-squared displacement (orange). Inset: Number of monomers per one cell of the primitive tube, N_e , as a function of the spacing a (symbols). The fitted power-law (green dashed) has the form $N_e = 0.85 a^{1.19}$.

$$D \sim v N_e b^2. \quad (5)$$

Comparing the latter relation with D computed from the long-time limit of the mean-square displacement of the center of mass, $D = \lim_{t \rightarrow \infty} g_3(t)/6t$, is a direct way to check the result (3). Such a confirmation is indeed independent as the long-time limit dynamics is juxtaposed with the contour velocity v that is extracted from short time scales as explained above. The number of monomers per cell, $N_e(a)$, that we needed for the comparison, has been measured independently from the primitive path length as $N_e = N/(L/a)$. The inset of Fig. 2 shows how N_e depends on the lattice spacing. For equilibrium polymers with no excluded volume, the polymer follows random-walk statistics $a \sim b N_e^{1/2}$, and in the case with excluded volume $a \sim b N_e^\nu$, where ν crosses over from 1 at small scales (rigid rod) to 0.588 (self-avoiding walk) at large. We find our data to fit well with $\nu \simeq 0.84$.

In Fig. 2, we plot together $v b^2 N_e$ and D as a function of $1/a$ and find an agreement without any fitting parameters. The scaling estimate $D \sim v N_e b^2$ captures well the non-monotonic behavior of D at high a^{-1} : initial increase of D with a^{-1} and then decrease of D from $a^{-1} = 1/4$ to $a^{-1} = 1/3$. Both facts result from the fine interplay of the finite size changes of v and N_e as a function of a . Recall that while v increases with $1/a$, the entanglement length N_e decreases with $1/a$. The regime of high $a \geq 13$ (slower drifts) is not captured by the scaling estimate accurately suggesting that either the diffusion along the primitive tube or transverse displacements start to play a role. As we show in Appendix Sec. B 3, all our systems are purely drift-dominated and the leveling off of the D at large a is rather caused by the transverse relaxation.

Our result (5) also agrees with the known expression for diffusion coefficient of a particle performing a persistent random walk with velocity $\tilde{v}$ that randomizes on timescale

$\tilde{\tau}$, i.e., $D_{\text{PRW}} \sim \tilde{v}^2 \tilde{\tau}$. We can view the motion of the center of mass of the chain as such a particle, where its velocity direction decorrelates in time $\tilde{\tau} = \tau_{\text{relax}} \sim (N/N_e)/v$. The velocity of the center of mass is, however, not equal to v , as that is a *contour* velocity of the chain and the center of mass is not in general bound to move along the contour. The center of mass moves the distance $R \sim bN^{1/2}$ in time $\tilde{\tau}$ and hence its velocity is $\tilde{v} \sim R/\tilde{\tau} \sim bN_e v/N^{1/2}$. As a result, the diffusion coefficient is $D_{\text{PRW}} \sim vN_e b^2$ consistent with (5) and our simulations. The distinction between the contour velocity and the velocity of the center of mass also illustrates that the dynamic characteristics of the center of mass and a chosen monomer can differ significantly on timescales shorter than τ_{relax} . In contrast to the center of mass, a monomer has to move along the chain contour around the obstacles. In what follows, we explore dynamic features of both in detail.

B. Segmental superdiffusive dynamics is coupled to the chain conformation

The entropic drift generates a distinctive dynamics of segments or their center of mass, relevant and accessible also experimentally in the context of chromatin fiber in living cells. Given the isotropy of the active entropic

pulling, the center of mass performs diffusion on long timescales ($t > \tau_{\text{relax}}$). However, below the relaxation time, the center of mass and the monomer dynamics can exhibit both subdiffusive and superdiffusive behaviors, depending on the activity of the segment and the examined length and timescales. In rough terms, on small length scales and short times the dynamics is Rouse-like (subdiffusive for a monomer), although modulated by the higher effective temperature. On longer scales, the entropic pulling starts to play a role and a superdiffusive regime emerges. A monomer performs ballistic motion over a curvilinear trajectory coupling the static conformational exponent with the dynamic one characterizing the mean-squared displacement, before the diffusive regime settles at long times.

To see the general trends discussed above in more detail, we measure the mean-square displacements $g_1(t)$ of the central monomer of the hot and cold segments respectively as in (A4). In the long-time limit g_1/t plateaus for all a and both segment types, signifying ordinary diffusion [Fig. 3(a)]. At short times the hot segments in systems with large a are subdiffusive, consistent with Rouse $g_1 \sim t^{x_1}$ with $x_1 = 0.5$ [Figs. 3(a) and 3(b)]. The active dynamics sets on when the hot segment explores several cells and, as a result, the cold segment is pulled and exhibits superdiffusive dynamics [$x_1 > 1$ in $g_1(t) \sim t^{x_1}$]. In contrast to

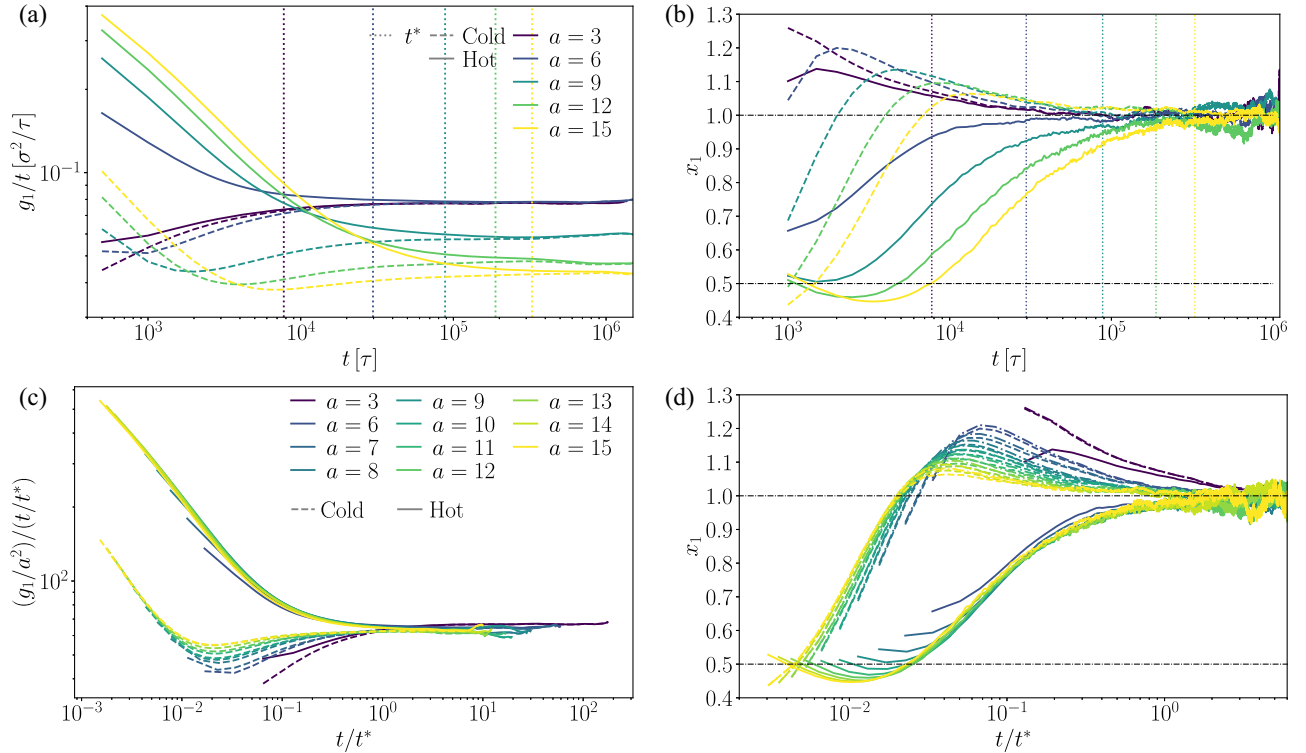


FIG. 3. (a) Mean square displacement divided by time $g_1(t)/t$ of a monomer in the middle of a hot (cold) segment, marked by solid (dashed) line, as a function of time for different lattice spacing a (colors). (b) Effective scaling exponent $x_1 = d \ln(g_1)/d \ln(t)$ of the mean squared displacement. Dot-dashed lines highlight exponents 1/2 (Rouse) and 1 (diffusion). (c), (d) The same data as in (a) and (b), respectively, normalized by a^2 and t^* (see text for definition, dotted) as marked in the axes labels. For clarity, fewer systems are shown in the non-normalized [(a), (b)] version.

systems with large a , the system with $a = 3$ is superdiffusive already at the smallest time resolution 500τ . This is true not only for the cold segment as expected, but also for the hot segment. The latter is caused by the fact that it is the chain tip that creates the entropic pulling force, while for such a small mesh size, the hot segment spans several cells and its midpoint already experiences the pulling of the tip. If we track the g_1 of the hot monomer at the very tip of the chain, we recover its subdiffusion (see Fig. 13). The case of $a = 3$ shows that an *internal* hot monomer can exhibit transient superdiffusion, which confirms that the pulling is generated at the tip and corroborates the entropic origin of the directionality.

We can recover a universal segmental chain dynamics if we coarse-grain the dynamics to the lattice scale, by normalizing the g_1 by a^2 and the timescale by a proper $t^*(a)$. The timescale $t^*(a)$ has to be larger than the relaxation time and such that in all systems the segments explore the same number of cells. We find a good collapse [Fig. 3(c)] of the g_1 curves if $g_1(t^*(a)) = (8a)^2$, which agrees with the number of cells ($\simeq 60$, Fig. 10) corresponding to the chain in the system with $a = 3$. To numerically evaluate t^* we used the above criterion for g_1 of the first cold monomer, but other monomer choices give essentially the same results, because the displacements coincide at long timescales as the monomers belong to the same chain.

The superdiffusive regime $x_1(t) > 1$ [Figs. 3(b) and 3(d)] stems from the fact that the monomer moves ballistically along the delineated (primitive) path, which exhibits non-ideal statistics ($2\nu > 1$) at these scales. In the artificial case of an infinitely long monofractal chain, characterized by 2ν independent of the scale s , the exponent x_1 would be a constant, $x_1 = 2\nu$, independent of the timescale. In reality, however, the superdiffusive exponent x_1 and its time-dependence are affected by three finite-scale effects: (i) the mesh length scale a , defining the onset of entropic pulling, (ii) the total chain length above which the system loses correlation and hence becomes diffusive and (iii) the scale dependence of the effective exponent $2\nu(s)$ that characterizes conformational features of the chain across the scales. Note that consideration of these effects can be important when evaluating measurements in real systems, such as chromatin, where the structural exponents are also scale-dependent and dynamical measurements are also typically available only in certain timescales [28].

To evaluate these effects in our simulations we compute the mean squared internal distance $d^2(s) = \langle (\mathbf{r}_i - \mathbf{r}_{i+s})^2 \rangle$, of a segment of length s , averaged over its position within the cold part of the polymer. From there the effective exponent $2\nu(s) = d \ln[d^2(s)] / d \ln s$ (Fig. 4). We compare $2\nu(s)$ with the exponent x_1 also computed for the cold monomers only. As both, $2\nu(s)$ and $x_1(t)$ depend on scale, we have to compare them on an equal footing. We focus on the maximal value that x_1 reaches in a particular system. As the superdiffusion emerges only on scales above a , the

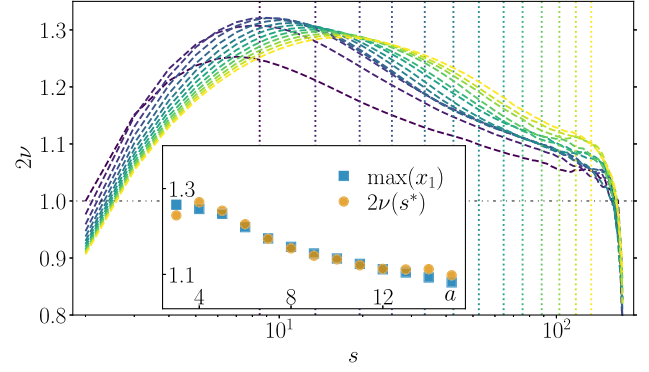


FIG. 4. The exponent 2ν of the cold segment as a function of the contour distance s . The vertical dotted lines are at s^* (see text). Inset: Comparison of the exponents $\max(x_1)$ and $2\nu(s^*)$. Note that the value of $\max(x_1)$ for $a = 3$ is underestimated because of the low sampling frequency.

exponent x_1 achieves its maximum only when the monomer moves some multiple $k \geq 1$ of cells. Therefore, we compare $\max(x_1)$ to $2\nu(s^*)$, where the latter is evaluated at the mean squared internal distance corresponding to the k multiple of the lattice constant a , i.e., s^* is defined by $d^2(s^*) = (ka)^2$. We find a good agreement for all the systems when $k = 3^{1/2}$ (Fig. 4 inset). More important than the actual fitted value of k is that other choices of k reproduce the dependence of $\max(x_1)$ on a up to a constant shift (Fig. 14), indicating the presented connection of 2ν and x_1 and that it is the finite length and time variation respectively generating the observed diversity of the exponents [Figs. 3(b) and 3(d)]. The connection of structural and dynamic exponents we present here complements and explains the missing superdiffusive regime for certain parameters reported in [45]. There, the dominating random walk statistics $\nu = 1/2$ of the primitive path implies just ordinary diffusion $x_1 = 2\nu = 1$.

C. Ballistic dynamics of the center of mass is limited by the chain length

While the monomers are bound to move along the curvilinear path of the chain, the center of mass of the chain is an abstract point that is not bound by the obstacles. As a result, its dynamic behavior and the corresponding exponents at the intermediate timescales can differ from those of the monomers.

In detail, the mean-squared displacement $g_3(t)$ of the center of mass of the whole chain, (A4), exhibits diffusion on long timescales ($t > \tau_{\text{relax}}$) as the primitive path of the chain loses orientational correlation. Note the diffusion sets on at about the same absolute time for all a [Figs. 5(a) and 5(b)], because the relaxation time $\tau_{\text{relax}} \sim (L/a)/v$ and both, the primitive path L/a (Fig. 10) and v are approximately proportional to a^{-1} . At early times, $g_3(t)$ shows the transition from sub-diffusive (reptation) to superdiffusive, which sets in at time $\hat{t}$ [Figs. 5(a) and 5(b)] when the hot segment initiates

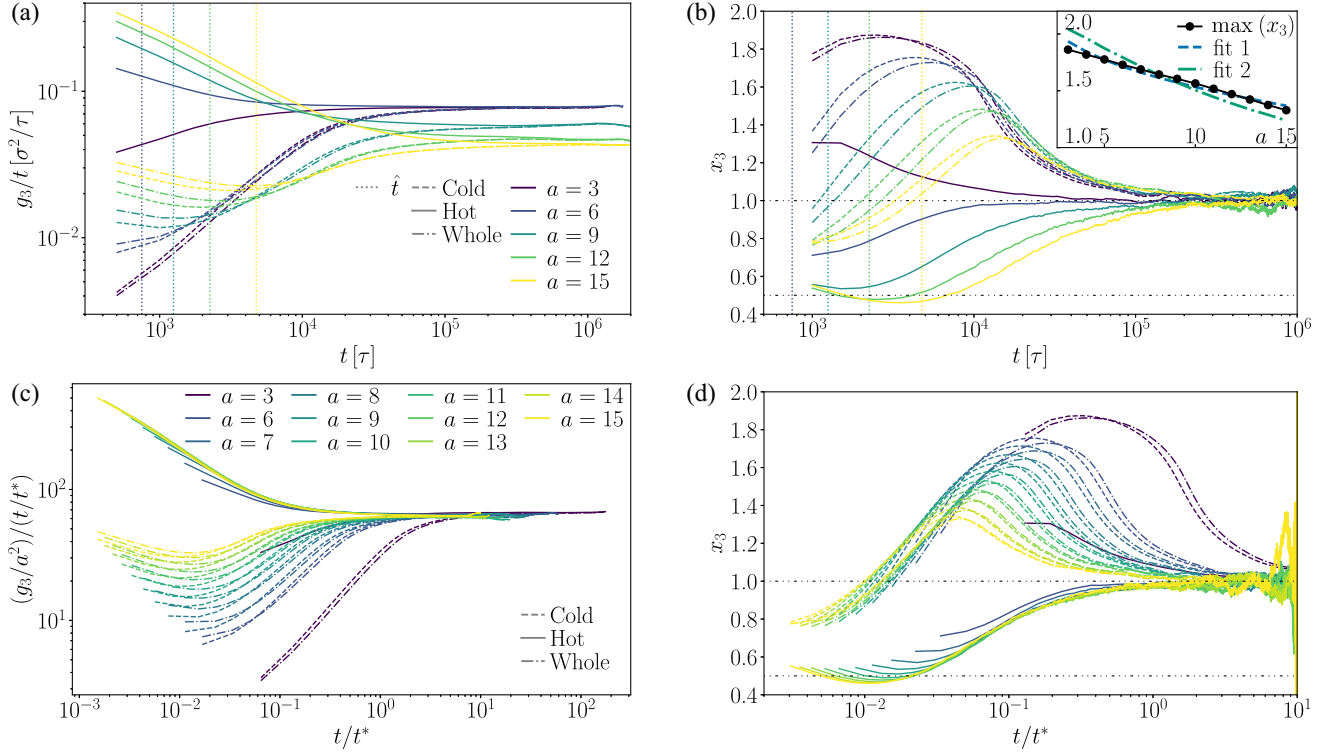


FIG. 5. (a) Mean square displacement divided by time $g_3(t)/t$ of a center of mass of the hot segment, cold segment or the whole chain (solid, dashed, or dot dashed) as a function of time for different lattice spacing a (colors). The vertical lines represent the time $\hat{t}$ when the hot segment's $g_1(\hat{t}) = 2.5a^2$ signifying the onset of the superdiffusion. (b) Effective scaling exponent $x_3 = d \ln(g_3)/d \ln(t)$ of the mean squared displacement. Dot-dashed lines highlight exponents 1/2 (Rouse dynamics for a segment, reptation for the total center of mass) and 1 (diffusion). Inset: The maximal exponent x_3 as a function of the lattice spacing a (symbols). Fit 1 is using (4) while fit 2 directly the measured values of $v(a)$ (see text in Sec. B 4). (c) and (d) Data normalized by a^2 and t^* (see text for definition) as in the axes labels. For clarity, fewer systems are shown in the non-normalized [(a),(b)] version.

the pulling by exploring a few cells ($g_1(\hat{t}) = 2.5a^2$). The highest exponent $x_3 = d \ln(g_3)/d \ln(t)$ is reached when the entropic pulling fully develops ($\approx 4\hat{t}$). In contrast to tangentially driven filaments where the center of mass typically exhibits ballistic motion, $x_3 = 2.0$, for $t < \tau_{\text{relax}}$ [15,46] we observe a -dependent exponents that arise from the finite chain size and transverse Rouse relaxation on scales smaller than a . Indeed, for small a , although each cell is explored by monomers quickly, the transverse fluctuations are strongly suppressed by the mesh, thereby the pulling resembles tangential driving. The primitive path of the chain spans tens of cells (Fig. 10 and Sec. B 4 in the Appendix), presenting long enough trajectory for the ballistic regime to develop. Conversely, large cells allow greater transverse fluctuations while the chain spans only several cells, limiting ballistic regime development. To see these effects we compare the time of the onset of pulling, i.e., when $g_3 \approx a^2$ and the time when the chain completely relaxes, by moving its own size, i.e., $g_3 \approx R_{ee}^2$. The corresponding time span, larger for smaller a , coincides with that when x_3 culminates (Fig. 15). Equivalently, one sees the limited time span by the higher $\hat{t}$ for higher a 's, while the time when the diffusion sets in is independent of [Figs. 5(a) and 5(b)].

Another way to illustrate the idea is when we recover the universal dynamics of the center of mass by coarse-graining to the level of the mesh [Figs. 5(c) and 5(d) as done for g_1 , by normalizing the time by t^* and g_3 by a^2]. The initial rise of the exponent $x_3 > 1$ exhibits overlap for systems with different a and lasts longer (and therefore reaches higher values) for systems of low a spanning more cells. More details on the crossover to the ballistic regime and its derivation are in the Appendix Sec. B 4.

Our results of x_3 can be compared to those of a tangentially driven chain explored in Ref. [45]. The work models the chain as a sequence of harmonic springs, each corresponding to one entanglement cell and hence the chain is essentially coarse-grained to the level of its primitive path. The different springs do not exhibit excluded volume and are all driven by a constant velocity in the contour direction. In contrast, our model has a finer resolution, excluded volume and is driven by the entropic force at the chain end. While the general trend of x_3 is consistent with our findings [inset of Fig. 5(b)] there exist fine differences that we attribute to the transverse fluctuations of the chain that were not considered in the analytical model [45] (see details in Appendix Sec. B 4).

D. Directional dynamics survives beyond mean field

All the above considerations were based on the mean-field-like assumption of the chain in a mesh of static obstacles representing the other chains in a melt system. Here we address the natural question if the emergent directionality survives when the inter-chain interactions and dynamics are considered explicitly. For that purpose, we simulated two full scale melts of $M = 500$ chains of length $N = 200$ at monomer density $0.85\sigma^{-3}$, where all chains within each simulation had the same stiffness, but the two simulations used different stiffness values. The stiffness controls the entanglement length N_e and the tube diameter a and, thereby, affects the mean-square displacement statistics. As explained in the appendix sec. A 1, we minimize the interfering effect of active-passive phase separation by the choice of $T_h = 2.0$. We compared the melt of fully flexible chains with one consisting of chains made semi-flexible by weak bending potential (A3). The corresponding entanglement lengths are $N_e = 70 \pm 5$ [47–49] and $N_e = 28 \pm 2$ [49] for the flexible and semi-flexible systems respectively. The associated tube diameters $a = 12.4$ and $a = 9.5$ are measured by the mean

squared internal distance corresponding to the entanglement length, i.e., $a = [d^2(N_e)]^{1/2}$. The results are summarized in Fig. 6, showing that the same directionality is present in the full melt simulations. Moreover, the result for the maximal superdiffusive exponent of the semiflexible chain $\max(x_3) = 1.5$ agrees almost perfectly with the predictions from our mean-field simulations and theory, where we found $\max(x_3) = 1.56$ for $a = 9$. The fully flexible melt system shows also clearly superdiffusive exponent $\max(x_3) = 1.22$, which is somewhat smaller than the mean-field result $\max(x_3) = 1.47$ for $a = 12$. We attribute this difference to the collective inter-chain effects, in particular the constraint release—the fact that in the melt the obstacles are formed by the other chains and hence are mobile [36]. In that case the relaxation is faster and the chains have shorter time to fully develop their superdiffusive dynamics and we observe smaller $\max(x_3)$. The constraint release effect is clearly stronger for less entangled chains, i.e. the fully flexible system. This is also supported by the fact that the time dependence of x_3 is more noisy for the fully flexible system than for the semi-flexible one and both are more noisy than the mean-field case.

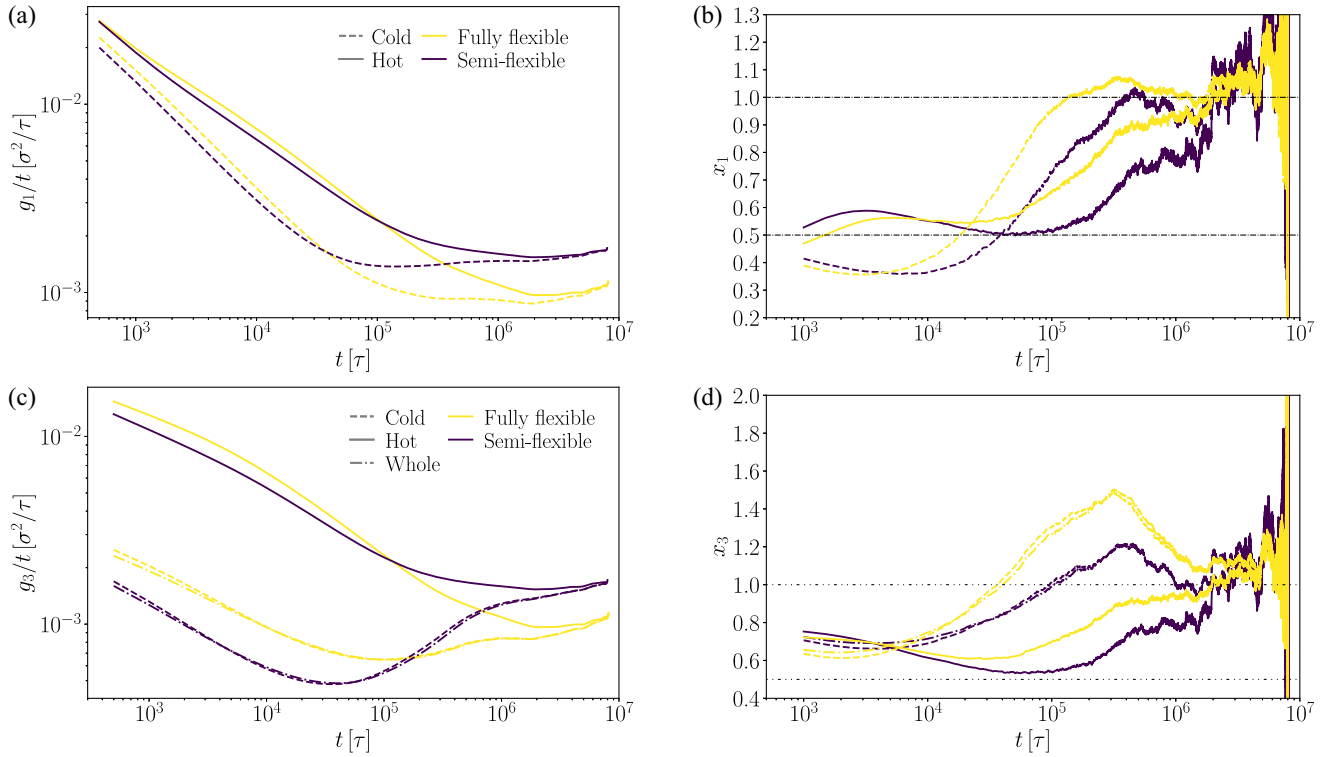


FIG. 6. (a) Mean square displacement divided by time $g_1(t)/t$ of the monomer in the middle of a hot (cold) segment, marked by solid (dashed) line segment as a function of time for different chain stiffness (colors). (b) Effective scaling exponent $x_1 = d \ln(g_1)/d \ln(t)$ of the mean squared displacement. Dot-dashed lines highlight exponents 1/2 (Rouse) and 1 (diffusion). (c) Mean square displacement divided by time $g_3(t)/t$ of the center of mass of the hot segment, cold segment or the whole chain (solid, dashed, or dot dashed) as a function of time for different chain stiffness (colors). (d) Effective scaling exponent $x_3 = d \ln(g_3)/d \ln(t)$ of the mean squared displacement. Dot-dashed lines highlight exponents 1/2 (Rouse dynamics for a segment, reptation for the total center of mass) and 1 (diffusion).

Another relevant interchain effect is the statistical nature of the entanglement mesh size. The computed N_e and a are averages that characterize the entanglements, but locally these quantities fluctuate, as there is no artificial mesh present in the melt simulations. Additionally, both are also affected by the fact that the hot segments have slightly lower stiffness than the rest of the chain. Similarly, partial active-passive microphase separation can affect the entanglement mesh statistics. We plan to address these finer effects in future, while at this stage, our melt simulations confirm the mean-field results.

III. DISCUSSION

Short-time directional correlations described for dumbbells can also occur for longer polymers in free space [25]. However, as we showed, in the presence of topological constraints the directional drift arising from the imbalance of the entropic forces governs the polymer behavior.

Besides the present directional dynamics, two-temperature polymer melts can exhibit also the emergence of the active topological glass (ATG) [10,11] and the active-passive (micro)phase separation [29,44]. While the precise interconnections of the three phenomena are not resolved, here we explain the physical origin of the directionality that is a necessary condition for the occurrence of ATG. In contrast to the ATG and active-passive phase separation, both of which occur only above a critical temperature contrast $\Delta T > \Delta T^*$, the pulling mechanism at hand occurs at any positive temperature contrast $\Delta T > 0$ as confirmed by $\lim_{v \rightarrow 0} \Delta T = 0$ in Fig. 1. Similarly, while the former phenomena are conditioned on a critical length of the hot segment, which depends on the temperature contrast, the present rectification mechanism, is at play as long as the chain possesses at least one hot monomer and the chain is larger than the lattice spacing. However, we expect that if the hot segment itself is smaller than the lattice spacing, the driving is caused by an effectively lower temperature contrast $\Delta T_{\text{eff},a} = T_{h,a} - T_c$, because the segmental dynamics is governed by the coarse-grained Langevin equation (1) on the scale a . On the scale of the cell the hot chain segment has the temperature $T_{h,a} = (T_h N_{h,a} + T_c N_{c,a}) / (N_{c,a} + N_{h,a})$, i.e., the number average temperature of the segment within the cell of size a . Therefore, $T_{h,a} = T_h$ only as long as the cell contains no cold monomers, $N_{c,a} = 0$.

The situation is more complex if the hot segment is located somewhere in between the chain extrema, which is a relevant case for chromatin model with heterogeneous activity (see next section). It has been shown analytically for a model *without* excluded volume or topological constraints that the activity induces effective looping of the active segment and effective attraction of the flanking passive regions [25]. Although the full model with the excluded or topological interaction is considered analytically inaccessible [25], here we show a way to treat it with

an effective theory. The entropic pulling of the intermediate hot segment competes with the stretching of the primitive path at the flanking ends, resulting in directional motion of the active loop. In that case, the chain will have the same dynamic features as the end-driven segment, characterized by the enhancement of x_1 beyond its passive values on intermediate timescales [10].

Notably, the observed superdiffusive regime in the melt, as well as in the simulations with static obstacles, occurs on relatively large timescales governed by the chain and entanglement scales. Interestingly, other models, for example chains with persistent driving [50], observe the regime of enhanced (super)diffusion between two sub-diffusive regimes at shorter timescales. These differences arise not only from the absence of confinement in our case, but also from the distinct mechanism of the superdiffusion that arise only after the hot segment can explore the entanglement scale. Such differences can be crucial in establishing, which mechanisms take place in experimental and biological systems such as chromatin.

A. Comparison to chromatin dynamics

Conformational data on chromatin do not reliably distinguish passive (segmental segregation due to distinct interactions) from active (due to different activity levels) mechanisms of chromatin folding, hence the dynamics is the decisive aspect to discriminate the mechanisms [25]. Here we compare our results with the available data on chromatin dynamics, assess what experimental challenges must be addressed to determine the *in vivo* relevance of these phenomena, and offer specific experimental predictions to guide future investigations.

Chromatin dynamics, typically measured by tracking the trajectories of fluorescent probes anchored to specific or nonspecific chromatin segments, is generally subdiffusive as expected for dense polymer solutions. However, several works recognized that active processes are manifested in chromatin coherent motion [21,51], enhanced diffusion or even superdiffusive behavior [52–56], but a diverse behavior upon transcriptional activation has been observed [57]. The heterogeneous dynamics likely arises from the different active processes and the heterogeneity of chromatin environment at different spatial and temporal scales. To assess if the active entropic pulling is present in chromatin we focus on the hallmarks of the phenomenon: the active dynamics (i) increases the mean-squared displacement exponents x_1 and x_3 beyond their equilibrium values; (ii) is present on scale comparable to, or larger than the entanglement mesh size a ; (iii) depends on the local compaction through a ; and (iv) can exhibit correlated displacements of the flanking segments next to an active segment.

The entanglement length scale $L_e = N_e B$, where B is the linear chromatin fiber density, and the related mesh size a are difficult to estimate for chromatin and can vary greatly, yet for human nuclei, both seem to be around

$L_e \simeq 0.1\text{--}0.5\ \mu\text{m} \simeq a$ [28]. The fact that $L_e \simeq a$ is the consequence of the chromatin fiber being relatively rigid on these scales as signified by the Kuhn segment of the chromatin fiber to be around $0.3\ \mu\text{m}$ [28]. Hence, the signatures of active entropic pulling can be expected on the scales above $0.1\ \mu\text{m}$. Note that these scales roughly interfere with the loop extrusion process [27] involving cohesin motors extruding loops of $\simeq 100\text{--}200\ \text{kbp}$ corresponding to about a $1\text{--}2\ \mu\text{m}$ contour length [28].

As conditions and activity are locus specific, the effective dynamic exponents x_1 are as well. The works [53–55] measured distributions of the dynamic exponent, based on the loci ensemble, with a majority of chromatin exhibiting subdiffusion but a subpopulation with superdiffusive behavior $x_1 > 1$. The sampled timescales are typically between 1 and hundreds of seconds corresponding to displacements of about $0.1\text{--}1\ \mu\text{m}$, which are in the right ballpark $\simeq a$. Different strategies are adopted to classify the heterogeneity. While the method of Shaban *et al.* [55] can classify the MSDs based on their spatial location, the other approach is to focus on specific chromatin sequences, such as telomeres (end regions) [53,54], euchromatic or heterochromatic sequences [1] or specifically ones that can be transcriptionally activated [23,57]. The observations show that euchromatic regions display larger MSD amplitudes and higher slopes in comparison to heterochromatin [1]. Notably, the global assessment of [55] shows that the MSD amplitudes are typically reduced upon activation, but the exponents (extracted as a single power-law fit of the data) are enhanced even beyond the diffusion limit with $x_1 \simeq 1.5$. A more detailed view on the level of single genes [57], shows two distinct regimes of MSD with the crossover time at about 10 s. At early times, below 10 s, the MSD amplitude is larger for activated genes than for the inactive genes but the slope of the former is lower (no quantitative fitting values reported) at longer times and hence the MSD of the passive genes dominates eventually. In contrast, the work [23] shows an increase of the effective slope from about 0.5 to 0.7 in the fitted time range of $0.1\text{--}10\ \text{s}$, where, however, the fit of the form $g_1 \sim A + Bt^{x_1}$ was used as the data was not a single power-law. Notably, the exponents depend on the local chromatin compaction. Although the exponents (or their time behavior) cannot be measured very accurately on the sample, the work shows the important differences in the active or inactive gene dynamics and their dependence on the local properties. Our work predicts the exponent of the active segment x_1 to reach the value $x_1 \simeq 0.7$ when the displacements are about $2a/3$ (Fig. 3) which agrees with the reported displacements on the order of $0.2\ \mu\text{m} \simeq a$ [23]. These results together illustrate that the effects are in the correct and measurable range, but a convincing proof requires to address several theoretical and experimental challenges that we list below.

The fitting form of the MSD should take into consideration the possibility of two distinct regimes, because, as

we see from our simulations too, the type of dynamics, characterized by the exponent x_1 , varies on the observable scales. The experimental measurements should focus on the exponents rather than on the prefactors. The prefactors of the fits, such as effective (sub)diffusion constants are more sensitive to friction, that can change, e.g., due to bulky motors attached (RNA polymerase). In contrast to the prefactors, the exponents are more stable [55] and can reveal the governing physical mechanisms. While our model is insensitive to the source of the active noise, the measurements complemented with biochemical interventions allow to decipher the origin of the activity such as loop extrusion [58], transcription, remodeling, etc., and the connection of the dynamics to structural features, such as lamin [53,54] or actin [52]. Clearly, the statistics can be improved by measuring a specific locus in a number of cells simultaneously, but the transcriptional activity can occur in bursts with silent periods and hence even with synchronization, the resulting MSDs are then a statistical mixture of both behaviors [23]. Also for these reasons, time averaging should be applied with caution as it might obscure finer details, such as superdiffusive bursts or other heterogeneity of the data [58].

Most studies focus on the MSD, which is the temporal autocorrelation function of the locus position, but a rich information is present in the cross-correlation function of distinct loci. The seminal work of Zidovska *et al.* [21] studied also the spatial correlations across the nucleus by nonspecifically, fluorescently tagging the histones. In contrast, here we suggest to focus on the correlations of loci spread *along* the studied segment. Our work offers a number of predictions for such an experiment based on Fig. 3: (i) distinct exponents for the activated (transcribed or hot) locus and the flanking (passive or cold) segments at a given time; (ii) lag time between the crossover to diffusion of the active segment and the increase of the passive exponent; and (iii) initial anticorrelation of the displacements of the flanking segments, as they are pulled towards each other by the actively driven segment, followed by correlated motion as they form the doubly folded segment driven further. Such observations should also be examined in the context of the work by Goychuk *et al.* [19], who study a free chain with spatially and temporally heterogeneous activity. In that work, depending on the temporal pattern of activity, its persistence time and the chain's bending flexibility, the chain can exhibit distinct folding regimes and correlated motion of distant loci. Such effects arise from the propagation of the active excitations by tension along the chain. Further set of predictions of our model concern the spatial variability of the dynamics arising from the heterogeneity of the underlying topological mesh of the chromatin due to varying density and stiffness of the fibers. A first step in this direction was the observation of the background chromatin motion in addition to the locus of interest in [23]. To interpret the results in

the context of our theory we would need additional understanding of the connection of the mesh size a and the chromatin properties that is so far lacking. An interesting future step is to investigate the impact of the hydrodynamic interactions that we neglect due to the typical screening effects in polymer melts in equilibrium. Yet, the local flow field around the pulled segment might lead to local coherent motion [21].

B. Effects on artificial active polymer melt

Polymers with heterogeneous fluctuations can be created artificially, for example, by coupling selected chain segment to a randomly fluctuating external field [10]. Polymer solutions and melts of such artificial polymers then can be used to test experimentally the predictions of our work. For example, the presented active dynamics impacts the viscosity. We expect the scaling of the viscosity with the polymer length to follow that of the tangentially active polymers $\eta \sim N^2$ [8], however, for a different reason. In the case of polymers tangentially driven at every monomer, the scaling arises from linear dependency of both, the relaxation time $\tau_{\text{relax}} \sim N$ [9], and the corresponding energy density of the active forces $G_0 \sim N$ [8], giving $\eta \sim G_0 \tau_{\text{relax}} \sim N^2$. In the case of two-temperature polymers, the $\tau_{\text{relax}} \sim \Gamma(L/a)/v \sim N^2$, while G_0 would not scale with length as the force is generated at the tip. These estimates rely however on the assumption that the entanglement effects are similar to the equilibrium case and independent of the driving, which might not be generally satisfied. Specifically, one assumption in this context that we use, and that was also used in directional reptation theory [16], is that the mesh size and the tube length are independent of the driving. For example, if the hot segments are located on both chain ends, the entropic force stretches the chain, which impacts the entanglement and dynamic properties [59]. As mentioned, the entanglement length scale is also sensitive to chain stiffness [47,49], which is certainly affected by the heterogeneous temperature along the polymer. These effects would hence deserve a detailed consideration. The superdiffusive dynamics exponents can also be tested experimentally on artificial polymers using fluorescent probes, differential dynamic microscopy [60], or direct microscopy of colloidal polymers. Whether these techniques can distinguish the tangentially driven polymers from the ones with heterogeneous fluctuations remains to be seen. A mere detection of superdiffusive dynamics would be a substantial progress as it was considered absent in two-temperature systems [50].

The present work explains how the broken translational symmetry due to (even temporary) topological constraints together with heterogeneous fluctuations generate directional motion of active-passive copolymers. We present how to understand these systems analytically using the language of equilibrium statistical mechanics expanded to several entropies associated to different thermostats.

The arising directional motion will likely affect steady state phases and the kinetics of both active-passive or genuine chemical copolymer melts. Exploring these directions further will bring deeper understanding of living chromatin and promises the development of functional active polymeric materials.

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J. S. conceived the study and formulated the theory. A. H., I. C., and J. S. ran the simulations and analyzed the data. All the authors interpreted the data; J. S. wrote the manuscript with contributions from all the authors.

DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

APPENDIX A: METHODS

1. MD model

We use the standard Kremer-Grest model with no bending rigidity. All beads interact with WCA potential

$$U_{\text{WCA}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] + \epsilon, \quad (\text{A1})$$

for $r < 2^{1/6}\sigma$ and 0 otherwise, where the scale σ represents the bead (monomer) diameter b . The mesh of obstacles is formed by immobile beads placed side by side (centers separated by σ) arranged in lines of a cubic grid with a beads per cell side. When referring to the grid cell size a as lattice spacing we use the units of σ , e.g., $a = 3$ means $a = 3\sigma$. The monomer beads interact with the mesh beads with the same potential (A1). The polymer connectivity is given by the finitely extensible nonlinear elastic potential

$$U_{\text{FENE}}(r) = -\frac{1}{2} r_{\text{max}}^2 K \log \left[1 - \left(\frac{r}{r_{\text{max}}} \right)^2 \right], \quad (\text{A2})$$

where $K = 30.0\epsilon/\sigma^2$ and $r_{\text{max}} = 1.5\sigma$, making the chains essentially uncrossable. The potential parameters, together with the bead mass $m = 1$ set the timescale $\tau = (m\sigma^2/\epsilon)^{1/2}$. Given the parameters this is of the same order as the monomer diffusion time $\tau_0 = b^2[\gamma/(k_B T)]$. All polymer beads are connected to their respective Langevin thermostat at temperature $T_{h/c}$, depending on if they are hot or cold, while both have a coupling (friction) parameter

$\gamma = 2/3\tau^{-1}$. All temperature values are in the units of ε . We use LAMMPS [61] to integrate the equations of motion with a timestep 0.005τ , while for the equilibration a larger time step of 0.01τ was used. To improve the statistics we run $n = 400$ independent copies of each system.

The melt simulation of fully flexible chains was using the same polymer model, while the semiflexible chains additionally had the bending potential

$$U_{\text{bending}}(\theta) = \kappa[1 - \cos(\theta)], \quad (\text{A3})$$

where θ is the angle of two subsequent bond vectors labeled in end-to-end sequential order and $\kappa = 1.5\varepsilon$. Initially, the melts were equilibrated at $T = 1.0$ for all the monomers and subsequently a head segment on each chain of length 25 monomers was made active by setting its thermostats to $T_h = 2.0$. As we showed [11], this lower value reduces the propensity to active-passive phase segregation by generating effective temperatures below the critical value for these chain lengths.

The mean-squared displacements were measured after a steady-state was reached. The latter was tracked by leaving-out the initial part of the whole trajectory and checking if the dynamics changes if a longer part of the trajectory was omitted.

The mean-squared displacement of a chain was calculated as the time average

$$g(t, t_{\text{tot}}) = \frac{1}{t_{\text{tot}} - t} \int_0^{t_{\text{tot}}-t} [\mathbf{R}(t' + t) - \mathbf{R}(t')]^2 dt', \quad (\text{A4})$$

where t_{tot} is the total simulation length after discarding the initial $10^5\tau$, for the simulations with mesh, and $5.10^5\tau$ for the melt simulations, in which the steady state is reached. The vector $\mathbf{R}$ represents the position of the center of mass of the chain (or its hot or cold segment) for the g_3 , or the position of a midpoint of the chain (or its hot or cold segment) for the g_1 . The presented mean-squared displacements are an average over all n chains.

APPENDIX B: SUPPORTING INFORMATION

1. Correlated noises in two-temperature dumbbell

A two-temperature dumbbell is described by two overdamped Langevin equations with temperature-dependent noise

$$\gamma_1 \dot{z}_1 = F(z_2 - z_1) + \sqrt{2\gamma_1 k_B T_1} \xi_1(t), \quad (\text{B1})$$

$$\gamma_2 \dot{z}_2 = -F(z_2 - z_1) + \sqrt{2\gamma_2 k_B T_2} \xi_2(t), \quad (\text{B2})$$

where we considered two different friction coefficients $\gamma_{1/2}$, $\xi_1(t)$ and $\xi_2(t)$ are independent white noises with zero mean $\langle \xi_i(t) \rangle = 0$, and unit variance $\langle \xi_i(t) \xi_j(t') \rangle = \delta_{ij} \delta(t - t')$, and F is the force of the spring.

The transformed variables, the relative distance $r = z_2 - z_1$, and the center of mass $R = (z_1 + z_2)/2$, then satisfy

$$\dot{r} = -F(r) \left(\frac{1}{\gamma_1} + \frac{1}{\gamma_2} \right) + \sqrt{\frac{2k_B T_2}{\gamma_2}} \xi_2(t) - \sqrt{\frac{2k_B T_1}{\gamma_1}} \xi_1(t), \quad (\text{B3})$$

$$\dot{R} = \frac{F(r)}{2} \left(\frac{1}{\gamma_1} - \frac{1}{\gamma_2} \right) + \frac{1}{2} \left[\sqrt{\frac{2k_B T_1}{\gamma_1}} \xi_1(t) + \sqrt{\frac{2k_B T_2}{\gamma_2}} \xi_2(t) \right]. \quad (\text{B4})$$

The noise terms for r and R are

$$\eta_r(t) = \sqrt{\frac{2k_B T_2}{\gamma_2}} \xi_2(t) - \sqrt{\frac{2k_B T_1}{\gamma_1}} \xi_1(t), \quad (\text{B5})$$

$$\eta_R(t) = \frac{1}{2} \left[\sqrt{\frac{2k_B T_1}{\gamma_1}} \xi_1(t) + \sqrt{\frac{2k_B T_2}{\gamma_2}} \xi_2(t) \right], \quad (\text{B6})$$

and their cross-correlation is

$$\langle \eta_r(t) \eta_R(t') \rangle = \left(\frac{k_B T_2}{\gamma_2} - \frac{k_B T_1}{\gamma_1} \right) \delta(t - t'). \quad (\text{B7})$$

Hence, even if the two friction coefficients $\gamma_1 = \gamma_2$ and there is no explicit r dependence in (B4), the noises remain coupled as long as $T_1 \neq T_2$.

2. Other chain properties

Other properties of the chains are presented in Figs. 7–10.

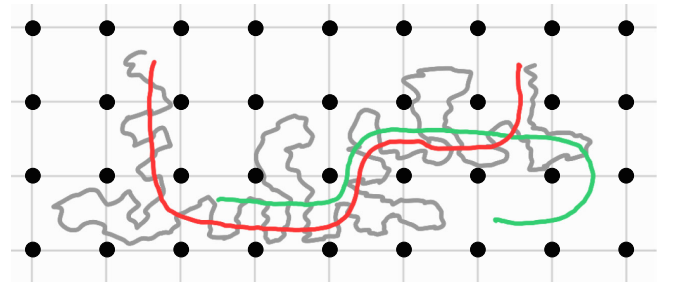


FIG. 7. Sketch of a chain conformation (gray) in a lattice of obstacles (black circles) with the corresponding primitive path (red). The cells (light gray squares) occupied by the primitive path correspond to the effective tube confining the chain. Number of monomers corresponding to one such cell is the entanglement length N_e . At later time, the chain moves and the corresponding primitive path (green) changes. A fraction of the tube that survives the change corresponds to those cells occupied by both, the red and the green paths.

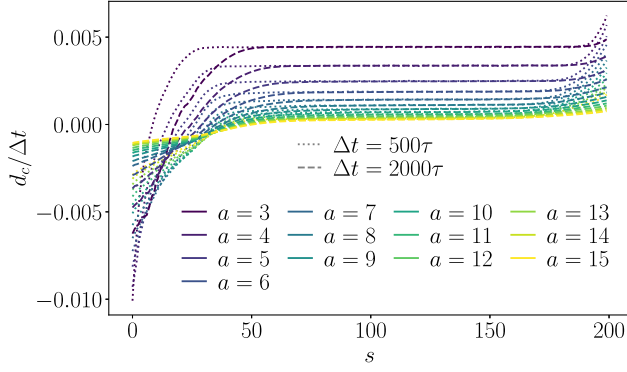


FIG. 8. Contour velocity $d_c/\Delta t$ as a function of contour coordinate s for systems of different a calculated from two different time lags $\Delta t = 500\tau$ (dotted) or $\Delta t = 2000\tau$ (dashed) for the systems with $T_h = 3.0$. The mean contour velocity is extracted as the mean over the region $s \in [75, 125]$ where $d_c/\Delta t$ is constant and hence independent of the different time lags. The decrease (negative) contour displacements of the hot segment ($s < 25$) and the upswing of the cold end are observation biases due to the end fluctuations. Only the fluctuating ends that do not enter the statistics of v are affected by different Δt .

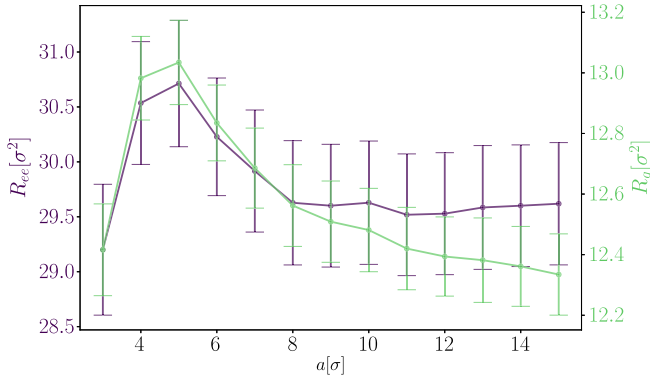


FIG. 9. Mean gyration radius R_g and the end-to-end distance R_{ee} of the systems as a function of the lattice spacing a . The mean performed over the steady state $t \geq 2.10^5\tau$. The error bars represent the standard deviation of the corresponding distribution. The ratio $R_{ee}^2/R_g^2 \simeq 6$ as expected for a random walk.

3. Drift or diffusion dominated relaxation and transverse displacements

As mentioned in the main paper the diffusion coefficient, as calculated from the drift-diffusion equation [45], for a tangentially driven polymer is in a dimensionless form given by

$$D = \frac{c}{6} \coth \frac{cZ}{2D_c}, \quad (\text{B8})$$

where Z is the number of lattice sites occupied by the chain. The dimensionless form is achieved by measuring the distances in lattice sites, and the time in the entanglement

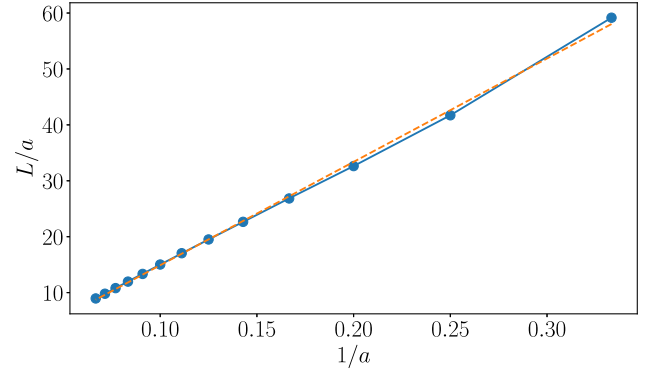


FIG. 10. The number of the lattice cells occupied by monomers of the chain without those involved in transverse excursion as a function of the inverse lattice spacing $1/a$. The quantity L/a also represents the mean length of the primitive path/tube over a . The mean (blue symbols) is taken over all conformations at times $> 10^5\tau$, spaced by $1.25 \times 10^4\tau$. The error bars representing the error of the mean are smaller than the symbol size. The linear function (dashed orange) is a guide to the eye showing a linear relation ($L/a \sim a^{-1}$).

time. Hence the velocity c is the number of lattice sites per entanglement time and the contour diffusion coefficient D_c is the number of lattice sites squared per entanglement time. The argument of the coth corresponds to the ratio $\tau_{\text{diffusion}}/\tau_{\text{drift}}$, i.e., the time for the center of the chain to move out of the initial tube purely by drift $\tau_{\text{drift}} = (Z/2)/c$ and by purely diffusion $\tau_{\text{diffusion}} = (Z/2)^2/D_c$. In our units these times are $\tau_{\text{drift}} = (N/2N_e)/v$ and $\tau_{\text{diffusion}} = (aN/2N_e)^2/D_c$, where D_c has the units of σ^2/τ and v just τ^{-1} . Therefore,

$$\tau_{\text{diffusion}}/\tau_{\text{drift}} = \frac{a^2 N v}{2N_e D_c} \equiv \text{Pe}, \quad (\text{B9})$$

is the Péclet number characterizing the ratio of the transport rate of advection to diffusion. For our values of the parameters and $D_c \simeq 1/200$ gives $\tau_{\text{diffusion}}/\tau_{\text{drift}}$ in the range from about $\text{Pe} = 50$ for $a = 15$ to about $\text{Pe} = 200$ for $a = 3$, meaning that our systems are drift dominated even for large a . This is confirmed by comparing the diffusion coefficient that we measured for the fully passive systems (Fig. 11) to the ones with the activity (Fig. 2) that are at least one order of magnitude larger. Moreover, we can use the Péclet number to find the parameter combinations for which the system is in the drift-dominated regime. For example, for a given drift velocity or temperature contrast ΔT one can compute the minimal polymer length for which the drift effects become relevant, by $\text{Pe} > 1$ in (B9).

In contrast to the tangentially driven chain on a lattice [45], our chains have the transverse freedom up to the cell size a . The transverse modes are responsible for the increased D in comparison to the scaling estimate vb^2N_e in cell sizes $a \geq 13$. Within the total relaxation time,

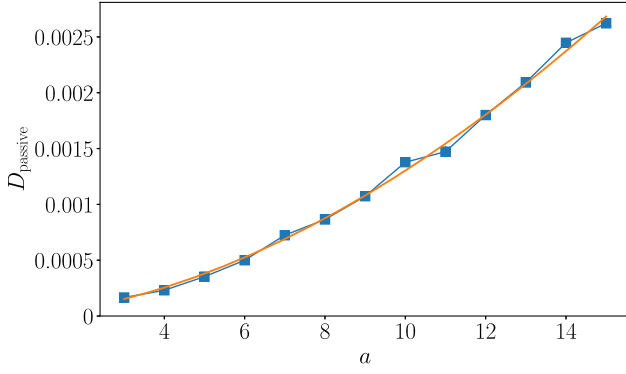


FIG. 11. Diffusion coefficient of the fully passive chains in equilibrium as a function of a measured as $D = g_3/6t$ (symbols). Orange line represents the fitted power-law $D = 2.155 \times 10^{-5} a^{1.78}$.

the chain is relaxed on the scale and therefore can move about one cell size in the transverse direction. If in each cell the chain moves about squared distance a^2 , and the displacements are independent, the center of mass squared displacement is about $a^2/(N/N_e)$. The scaling estimate of the diffusion coefficient $D \sim R^2/\tau_{\text{relax}}$ should be amended by such a term giving

$$D \sim \frac{R^2 + (a^2/(N/N_e))}{\tau_{\text{relax}}} = vb^2N_e \left[1 + \frac{a^2N_e}{b^2N^2} \right]. \quad (\text{B10})$$

The term in the square bracket hence represents a correction factor to the original scaling estimate vb^2N_e . To see whether it can fit the data, we perform a two parameter fit $A[1 + Ba^2N_e/b^2N^2]$ of $g_3/6t$. The overall pre-factor A and the factor B are used because the presented displacements are only scaling estimates and the displacements in different cells are not necessarily independent. As presented in Fig. 12 the fit works relatively well in the regime of large a .

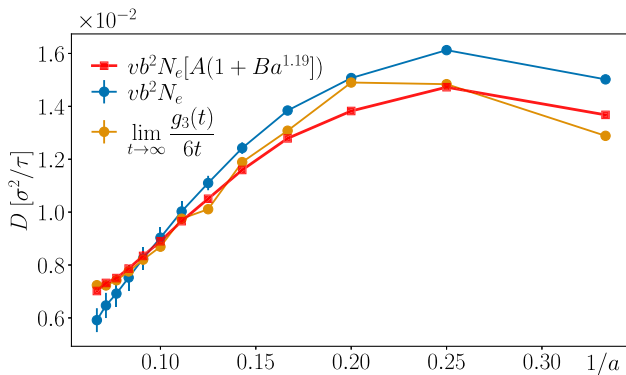


FIG. 12. Diffusion coefficient of the active system (orange), simple scaling estimate (blue), and the scaling estimate with the transverse displacement correction (B10) (red) where we used the fitted form of $N_e \sim a^{1.19}$ from the main paper.

4. More details on g_3 calculation and comparison with tangentially driven chains

Tejedor and Ramírez [45], considering a tangentially driven chain with constant drift on the tube level, explain the superdiffusion exponent x_3 as a crossover between the short time diffusion dominating over drift and long time diffusion above the chain scale that is governed by the drift. Based on the crossover times between the regimes and the value of g_3 they derive an approximate expression for $x_3 \simeq 0.657 \log(cZ^2) + 1.77$, where c is the drift in the lattice units per entanglement time $\tau_e = \tau_0 N_e^2$, with τ_0 being the monomer diffusion time, and $Z = N/N_e$ is the number of entanglements. In our units, this translates to $x_3 \simeq 0.657 \log(Av) + 1.77$, where $A = N\tau/\tau_0$ and the only dependence on the entanglement (lattice) scale comes from the velocity $v(a)$. We use this approximation to fit the data of the $\max(x_3)$ reached in the simulation for the given a , in two ways. In the first one, we used the expression (4) for $v(a)$, using A and a_{max} as the fitting parameters, and in the second one we use directly the measured values of v and using only A as a fitting parameter. Both fits capture the main trend [inset of Fig. 5(b)], but fail to reproduce the measured dependency details as highlighted by the disagreement of the fitted parameter $a_{\text{max}} \simeq 42$ with the one obtained from (4) $a_{\text{max}} \simeq 18$. We attribute these differences to explicit transverse fluctuations of the chain that speedup the chain relaxation (Sec. B 4) and were not considered in the analytical model [45], but further work is needed to confirm this hypothesis.

On timescales shorter than the relaxation time $t < \tau_{\text{relax}}$, the lattice prevents rotational relaxation of the chain on the scale above the mesh size. Therefore, as the entropic pulling is at play, the mean squared displacement of the center of mass $g_3(t)$ is similar to that of the tangentially driven polymers [46] exhibiting ballistic superdiffusion, but differs in the faster crossover to diffusional regime. Below we recapitulate the calculation of [46] and show where our system differs.

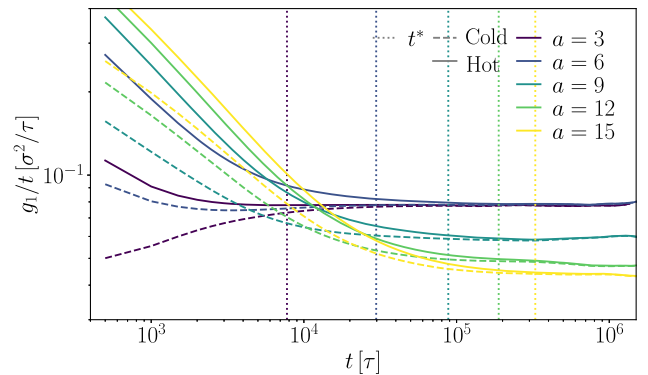


FIG. 13. Mean squared displacement over time g_1/t of the first monomer of the segment as a function of time. In contrast to the central monomer (see main text), the first monomer of the hot segment exhibits subdiffusive behavior at short times for all systems (even $a = 3$) as expected.

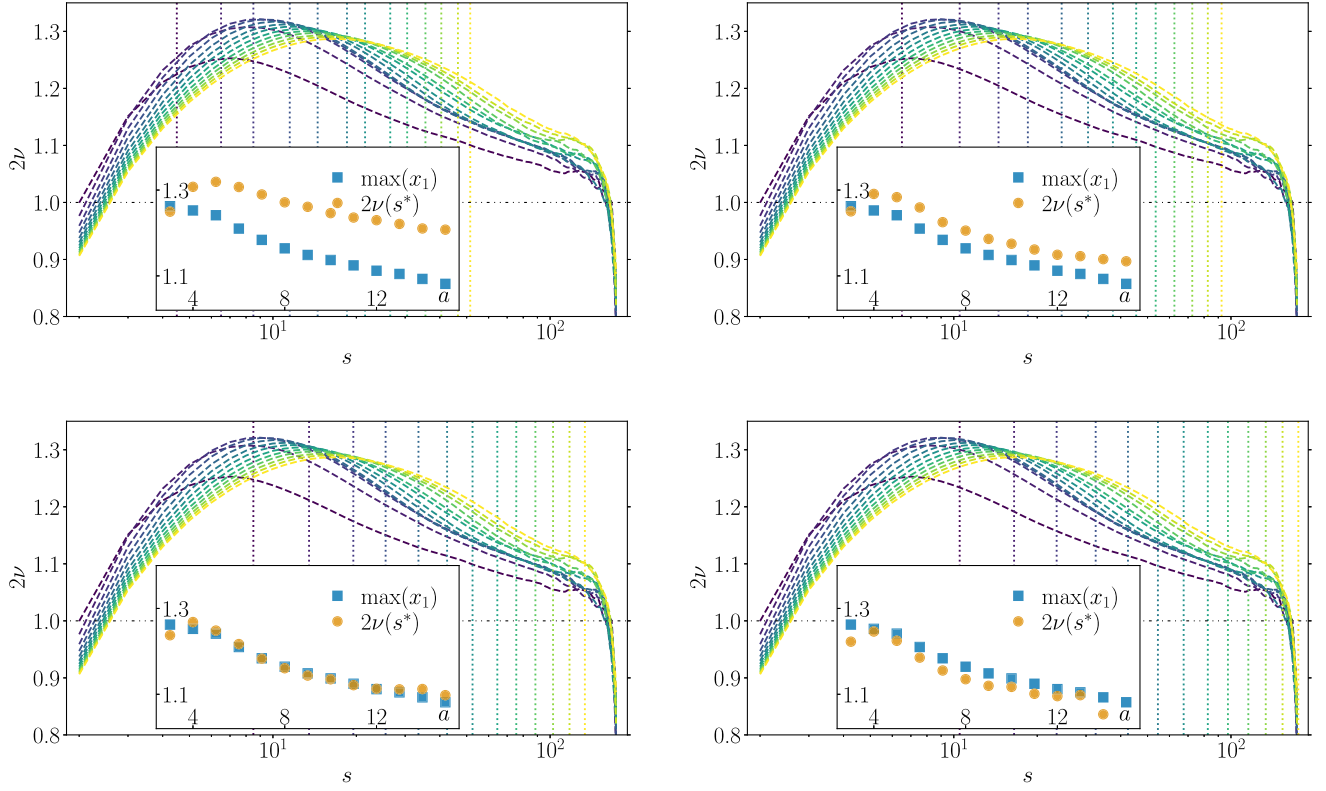


FIG. 14. The exponent 2ν of the cold segment as a function of the contour distance s . The vertical dotted lines are at s^* (see main text for details) extracted from when the means squared internal distance $d^2(s^*) = (ka)^2$. Here the top row corresponds to $k^2 = 1$ (left), $k^2 = 2$ (right), while the bottom row to $k^2 = 3$ (left—identical to the plot in the main paper) and $k^2 = 4$ (right). Insets: Comparison of the exponents $\max(x_1)$ and $2\nu(s^*)$.

The center of mass position vector $\mathbf{r}_{\text{cm}}(t) = N^{-1} \sum_{i=1}^N \mathbf{r}_i(t)$, with the monomer i position vector $\mathbf{r}_i$. As the polymer moves along its contour path, the displacement $\Delta \mathbf{r}_{\text{cm}}(t) = \mathbf{r}_{\text{cm}}(t) - \mathbf{r}_{\text{cm}}(0)$ can be expressed as

the sum of the end-to-end vectors $\mathbf{r}_e$ of the whole polymer at times t_i from $t_0 = 0$ to $t_k = t$ (see Fig. 16)

$$\Delta \mathbf{r}_{\text{cm}}(t) = N^{-1} \sum_i^k \mathbf{r}_e(t_i). \quad (\text{B11})$$

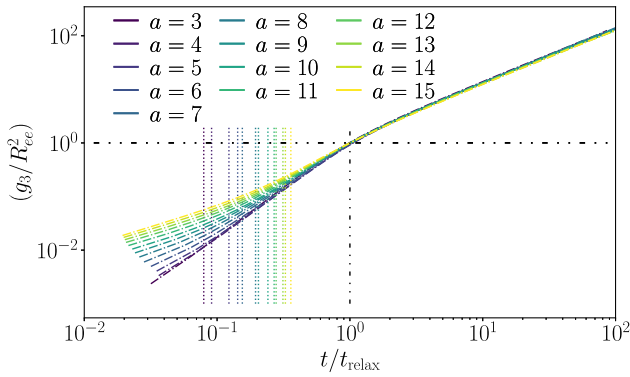


FIG. 15. Mean squared displacement of the chain's center of mass g_3 normalized by the mean squared end-to-end distance R_{ee}^2 as a function of time normalized by the time t_{relax} , computed here as $g_3(t_{\text{relax}}) = R_{ee}^2$. The vertical colored dotted lines show time when $g_3 = a^2$, while the vertical black dash-dotted line is at $t/t_{\text{relax}} = 1$. The time span between the vertical colored and the black line coincide with superdiffusive regime of g_3 . Longer superdiffusive regime is observed for systems with lower a .

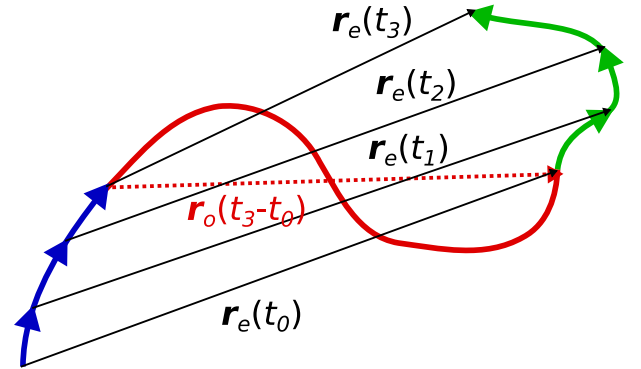


FIG. 16. The polymer path and its end-to-end vectors $\mathbf{r}_e(t_i)$ at different times t_i . The green part represents the path segment that the head of the polymer moved between t_0 and t_3 , while the blue one similarly, but for the polymer tail. The red part represents the segments of the conformation that overlaps between t_0 and t_3 and the red dashed line is the corresponding end-to-end vector $\mathbf{r}_o$.

Switching to continuous representation $\sum_i^k \rightarrow \tau_b^{-1} \int_0^t dt'$, where τ_b here is the unit of time (monomer to move its own size by drift: $b/v = \tau_b$), with b the monomer size, the $g_3(t)$ is then

$$g_3(t) = (N\tau_b)^{-2} \int_0^t \int_0^t \langle \mathbf{r}_e(t') \mathbf{r}_e(t'') \rangle dt' dt''. \quad (\text{B12})$$

At any time the end-to-end vector $\mathbf{r}_e(t_i)$ can be decomposed in the sum $\mathbf{b} + \mathbf{r} + \mathbf{g}$, where the terms correspond to the sections of blue, red, and green segments, respectively. As the newly traced segments of the polymer are uncorrelated to all the previous ones, i.e., $\langle \mathbf{g}\mathbf{r} \rangle = \langle \mathbf{g}\mathbf{b} \rangle = \langle \mathbf{r}\mathbf{b} \rangle = 0$ (Fig. 16), the only contribution surviving the averaging in (B12) would be the overlapping (red) segment terms $\langle \mathbf{r}_o^2(t'' - t') \rangle = \mathbf{r}_o^2(t'' - t')$. Therefore, we can express g_3 as

$$g_3(t) = (N\tau_b)^{-2} \int_0^t \int_0^t \mathbf{r}_o^2(t'' - t') dt' dt'', \quad (\text{B13})$$

where the time dependence of the $\mathbf{r}_o^2(t)$ is

$$\mathbf{r}_o^2(t) = b^2 \left[N - \frac{vt}{b} \right]. \quad (\text{B14})$$

Here b is the bond length and we assumed ideal chain statistics and uniform driving velocity. Then, integrating (B13) gives the ballistic regime $g_3 \sim t^2$ for $t < N\tau$ and the diffusive regime $g_3 \sim t$ for $t > N\tau$.

All the above holds because a tangentially driven chain follows closely its own path. In our case it holds true on the scales above a because significant rotational diffusion is suppressed by the obstacles. However, when the overlap segment $\mathbf{r}_o$ spans just one cell size it can relax by rotational diffusion before it is pulled out completely. The criterion is $\tau_a > \tau_e$, that is $a/v > \tau_0 N_e^2$, which for $v \simeq k_B \Delta T (a^{-1} - a_{\max}^{-1})/N\xi$ and $\tau_0 = b^2 k_B T_c / \xi$, gives the criterion in the form

$$a^2 \left(1 - \frac{a}{a_{\max}} \right) < R^2 \frac{T_c}{\Delta T}, \quad (\text{B15})$$

where we used $R^2 = Nb^2$. This inequality is satisfied for $a \ll a_{\max}$ (recall $a_{\max} \simeq R_g$) for not too strong temperature contrasts or when $a \simeq a_{\max}$. In either case, for the $\Delta T \simeq 2T_c$ that we use the chain is mostly relaxed on the scale of the cell. For small a , the chain spans many cells and therefore the crossover to ballistic regime can be observed, because the relaxation of $\mathbf{r}_o$ in the last cell does not affect the overall scaling. However, for large a there are only few cells and the relaxation is relevant. As the transverse modes are significant at the chain end (see Fig. 17) the integrand of (B12) we approximate as $\langle \mathbf{r}_e(t') \mathbf{r}_e(t'') \rangle \simeq \langle \mathbf{r}_o(t'') \mathbf{r}_o(t') \rangle$ and further by asserting that

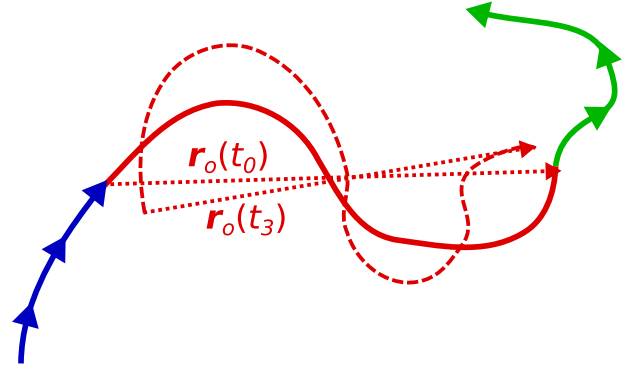


FIG. 17. The polymer between times t_0 and t_3 mostly follows the delineated path, but because of the transverse modes the overlapping (red) region is no longer perfectly correlated but can decorrelate.

$$\langle \mathbf{r}_o(t'') \mathbf{r}_o(t') \rangle = \langle \mathbf{r}_o^2(t'' - t') \rangle = b^2 \left[N - \frac{v(t'' - t')}{b} \right] \quad (\text{B16})$$

only for $t'' - t' < t_a$ and 0 otherwise, where $t_a = (N - N_e)b/v$ is the time for the active force to pull out the polymer from its tube completely up to the last cell. In other words we assume that as the chain is completely relaxed below the tube the scalar product of $\mathbf{r}_o$ is zero in the last cell, but stays ideal in all other cells, because of the limited transverse relaxation above a . Then Eq. (B13) becomes

$$g_3(t) = 2 \frac{b^2}{N^2 \tau^2} \int_0^{t-t_a} dt' \int_{t'}^{t_a+t} [N - v(t'' - t')/b] dt'' + 2 \frac{b^2}{N^2 \tau^2} \int_{t-t_a}^t dt' \int_{t'}^t [N - v(t'' - t')/b] dt''. \quad (\text{B17})$$

In the regime when $t_a < t < t_{\text{relax}} = N\tau = Nb/v$, which is the regime when the chain is yet to be pulled out from the last cell, the result is

$$g_3(t) = \frac{b^2}{N^2 \tau^2} \left[t_a \left(t - \frac{t_a}{2} \right) + \frac{v}{Nb} \left(-t^2 t_a + \frac{3}{2} t t_a^2 - \frac{2}{3} t_a^3 \right) \right]. \quad (\text{B18})$$

In this regime the first term in (B18) dominates (their ratio is roughly t_{relax}/t) and therefore $g_3 \sim t$. The mean-squared displacement crosses over from superdiffusion to ordinary diffusion even before the whole tube is vacated due to Rouse relaxation of the chain end on the scale of a .

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